

Customer Segmentation: Clustering and PCA

MKT 566 · Fall 2026

Davide Proserpio

USC Marshall School of Business

Before We Start

Housekeeping

- **Homework 2** (box office) due **Thursday, Oct. 8**
- **Group project**: proposal presentations on **Oct. 13 and 15**
- Today: **clustering** (k-means) and **PCA**. Thursday: **recommendation systems**

Proposal presentations: how we grade them

Oct. 13 and 15, 15 minutes per group. Send your slides by **Monday, Oct. 12**. The plan does not need to be finished: this is where you get feedback.

What we look at	What a good proposal does
The question	says clearly what you want to find out, and why a marketer would care
The data	says where your data comes from, and shows you can get it
The method	says how you will answer the question, with at least one tool from this class
The talk	clear slides, finishes on time, speaks to the room, handles questions well

Each of the four parts gets **10, 14, 16, 18, or 25 points**, from “needs improvement” to “excellent”, so the total is out of **100**. The proposal counts for **15%** of your group project grade.

A different kind of question

Every model so far had an **outcome** to predict: a listing's price, then whether it is a gem. Today there is no outcome. There are just customers, described by numbers, and one question: **who is similar to whom?**

What changes

- No Y , no coefficients, no stars: the algorithm proposes **groups**
- This is called **unsupervised** learning: nothing tells the algorithm what the right answer is

What stays

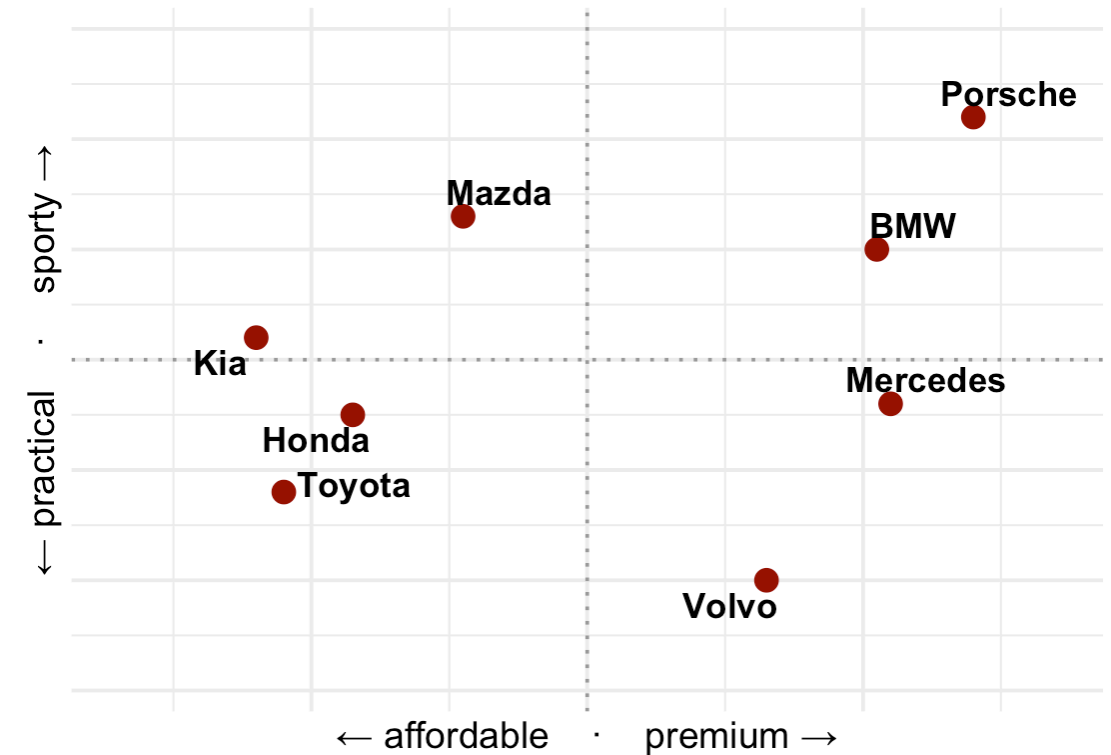
- The data is still rows and columns, and **you** choose the columns
- The output still means nothing until **you** interpret it
- Still **not causation**: clusters describe, they do not explain

Why marketers cluster (and map)

Grouping similar customers is **segmentation**, and almost everything in marketing is segment-specific:

- **Messaging**: bargain hunters get the discount email, service seekers get the concierge pitch
- **Product and pricing**: which features matter to which group, and who will pay for them
- **Budget**: which segments are growing, and where the next dollar goes
- **Positioning**: the same kind of data, mapped, shows who competes with whom and where the gaps are: the **perceptual map** on the right

The car market, as customers see it

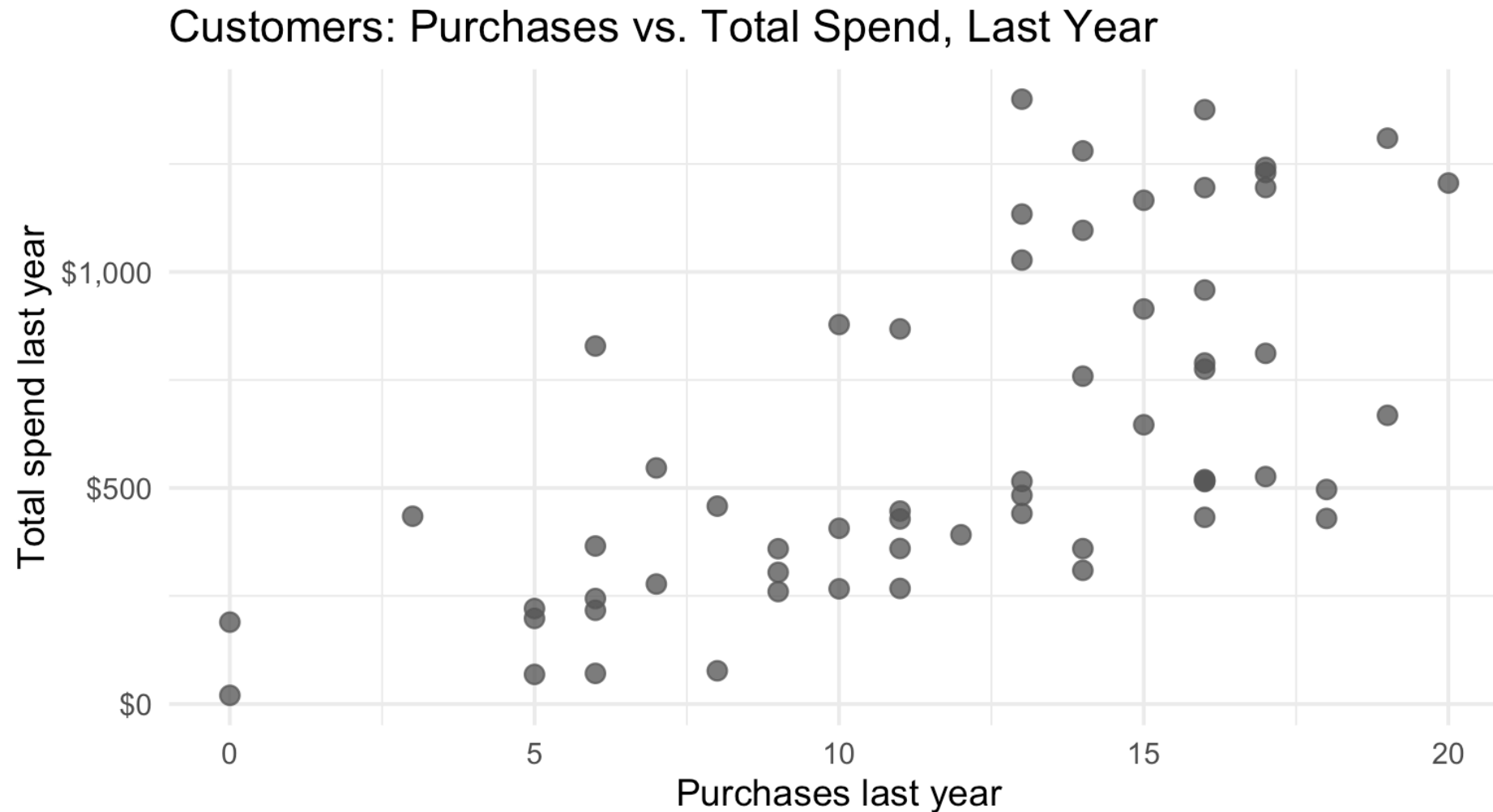


A sketch, not data. You will build a real one with PCA at the end of today.

Two tools today. **K-means** finds the groups. **PCA** draws the map you need to see them, and it powers the perceptual map.

Finding Groups: K-Means

Sixty customers, two numbers



Each dot is a customer: how often they bought, and how much they spent in total. How many groups do you see, and how would you describe each one?

The same chart, after k-means



- We gave k-means only the **numbers** and **k = 3**. Where your eyes hesitated, it drew the borders by **distance**, the same way every time. Dots near a border could go either way: normal with real customers
- The **X** marks each cluster's center. Read the centers and the groups get names: **occasional shoppers** (about 6 purchases, \$290), **frequent small-basket shoppers** (about 14 purchases, \$560), and **big spenders** (about 16 purchases, \$1,180)
- Eyes can handle 2 columns and 60 customers, not **6 columns and thousands of customers**. The algorithm can, with the same logic

How k-means works

You pick **k**, the number of clusters. The algorithm does the rest:

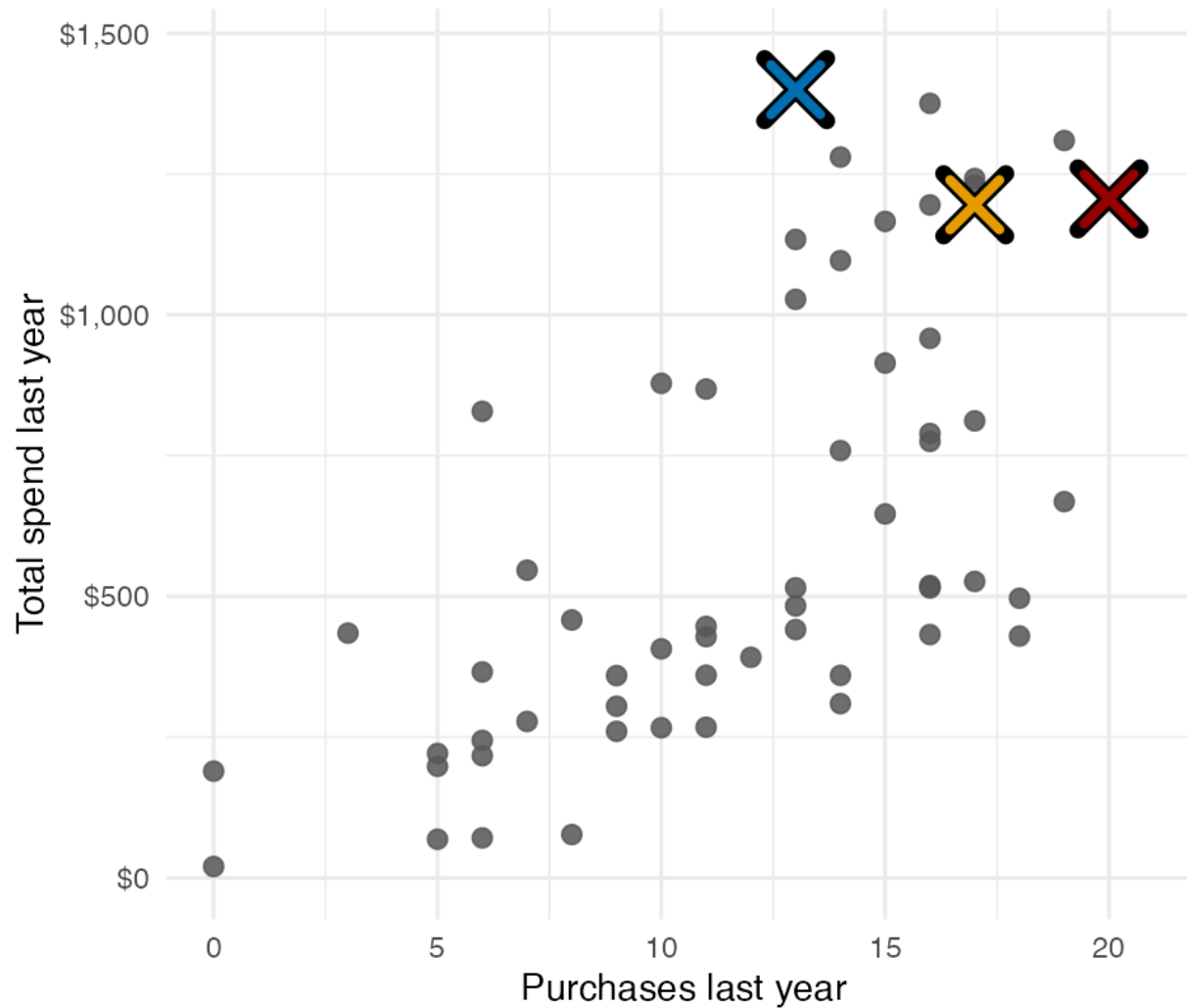
1. Place **k centers** at random
 2. **Assign** every customer to the nearest center
 3. **Move** each center to the middle of the customers assigned to it
 4. Repeat 2 and 3 until the centers stop moving
- On the last chart, that is: drop three **X**'s at random, color every dot by its nearest X, move each X to the middle of its color, repeat. The X's you saw are where they stopped
 - “Nearest” means plain **distance** between rows of numbers, so every column you feed it counts
 - The random start means results can vary from run to run. In practice, k-means is run many times (say, 20) from different starts, and the best result is kept

K-means never sees names, labels, or meanings. It groups rows of numbers by distance. Everything the clusters *mean* comes later, from you.

K-means, step by step

Step 1: drop 3 centers

anywhere, at random



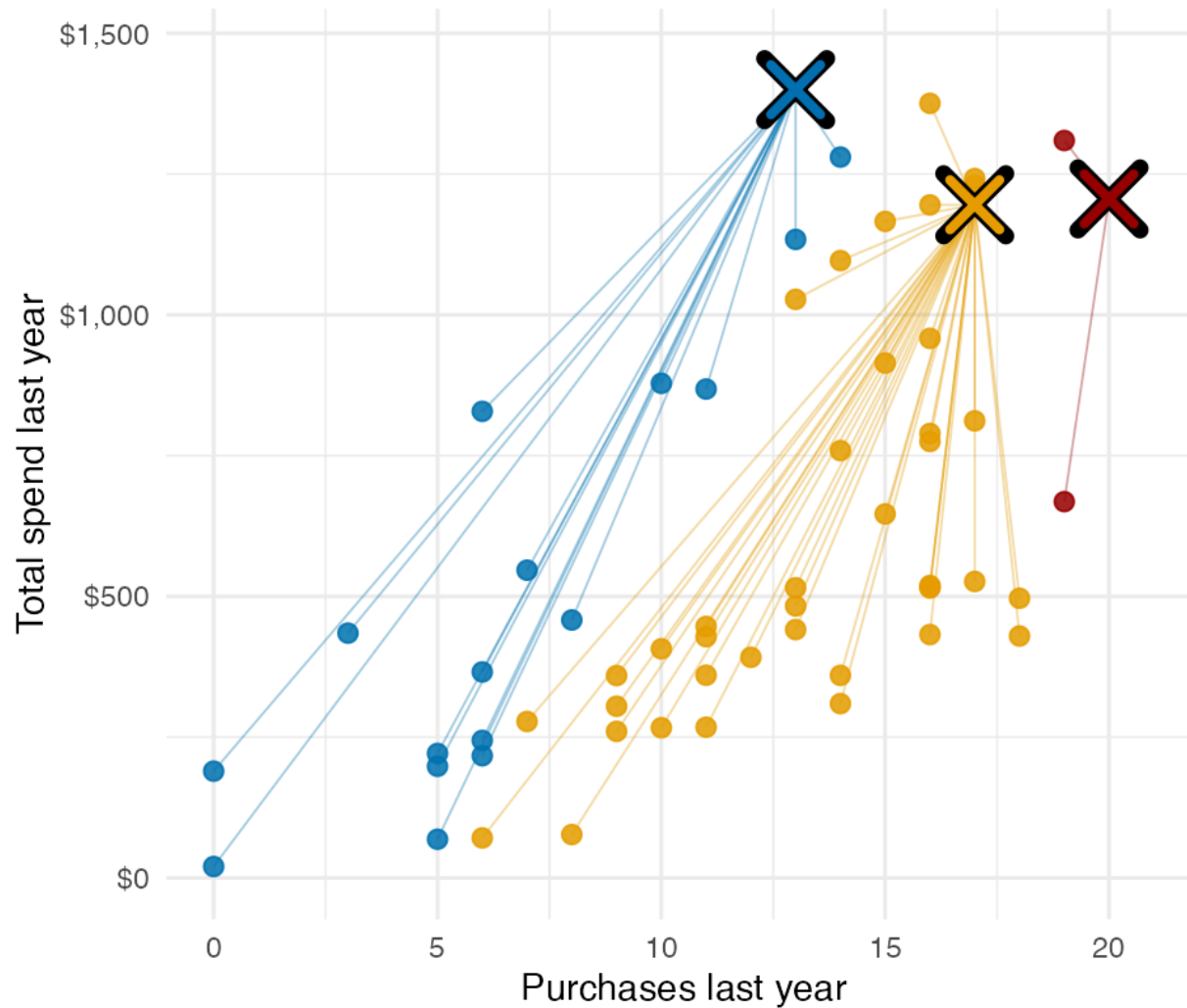
1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

K-means, step by step

Step 2: assign

each customer takes the color of its nearest X



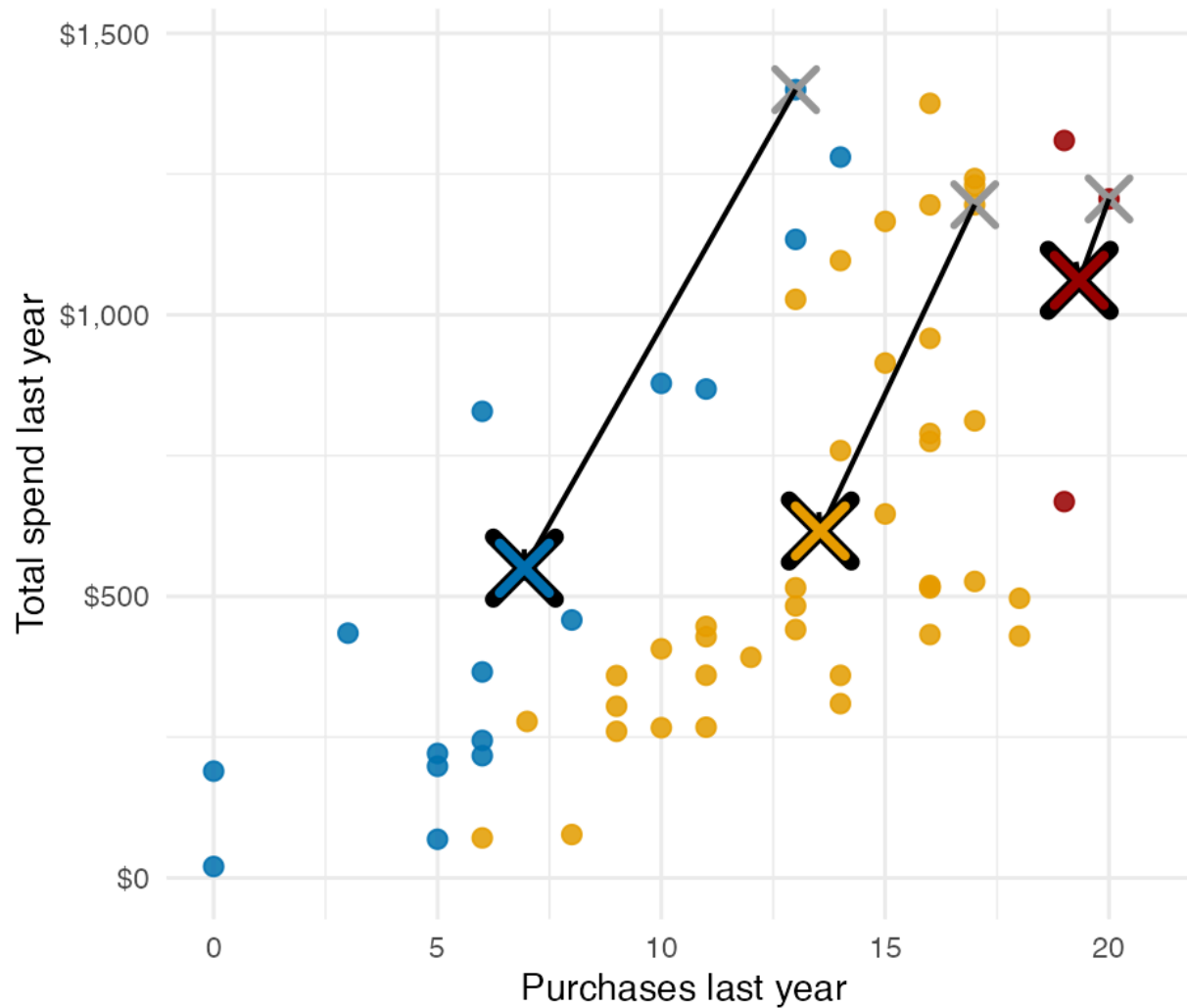
1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

K-means, step by step

Step 3: move

each X moves to the middle of its color



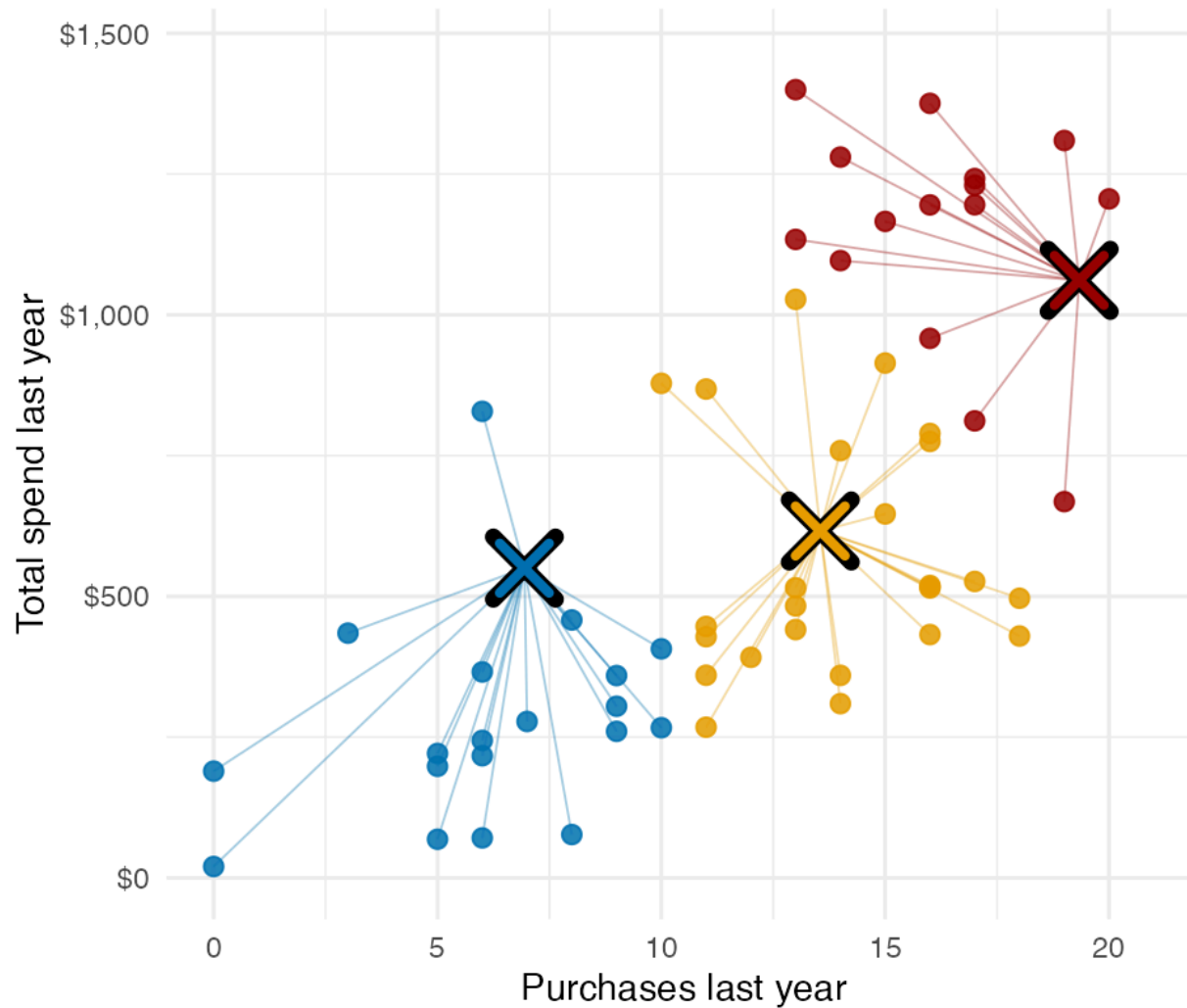
1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

K-means, step by step

Step 2 again: assign

22 customers switch color



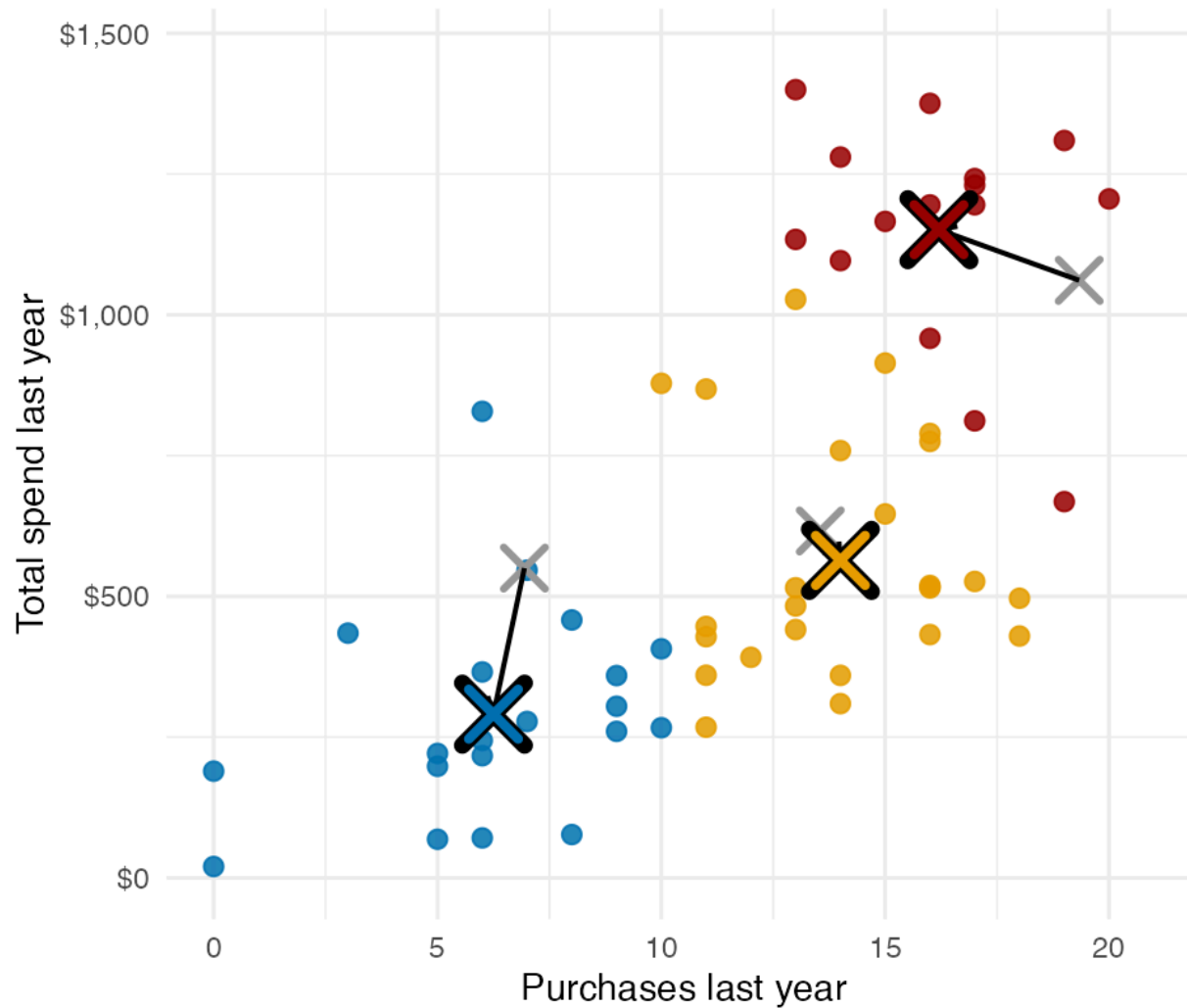
1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

K-means, step by step

Step 3 again: move

each X moves to the middle of its color



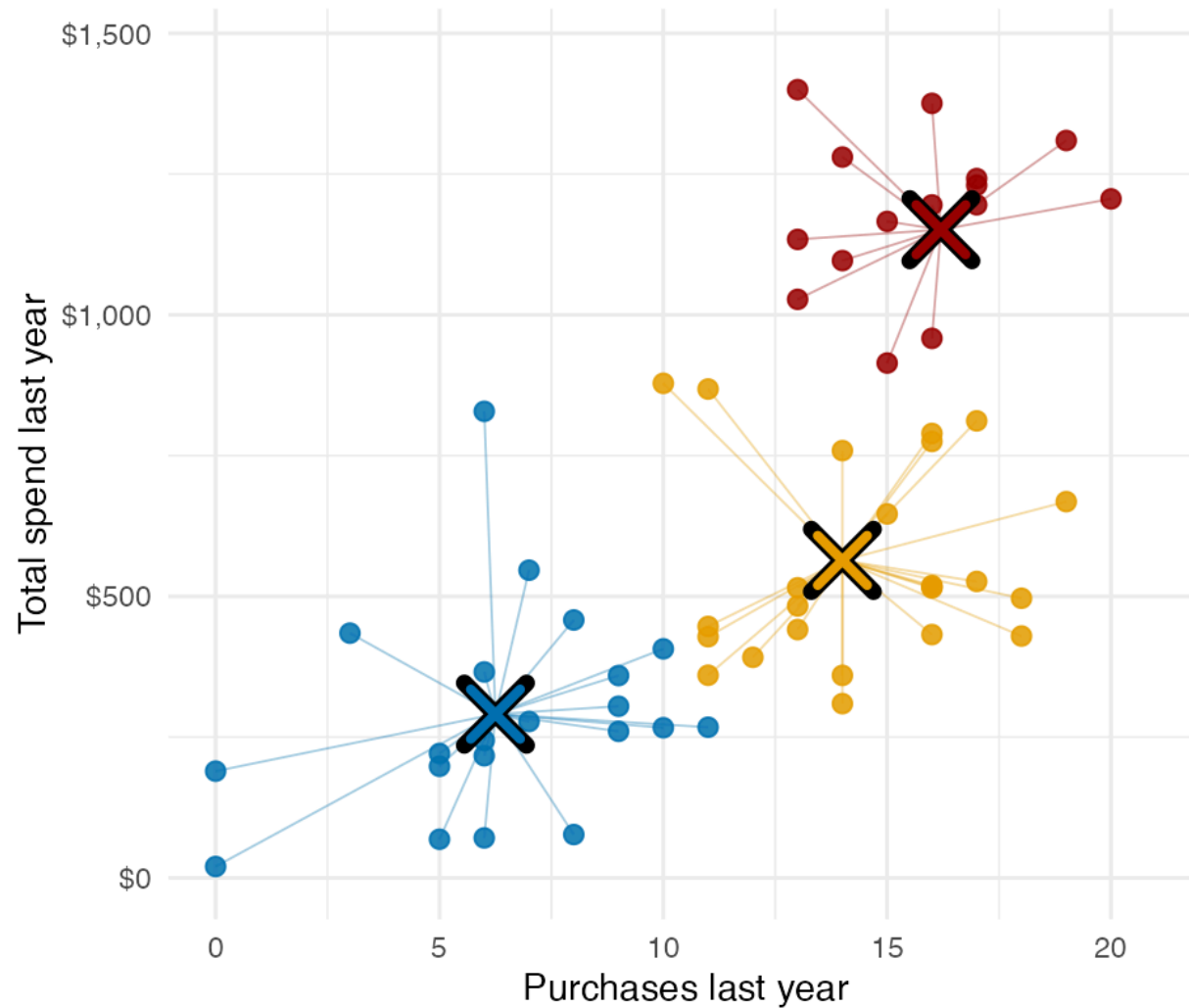
1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

K-means, step by step

Step 2 again: assign

5 customers switch color



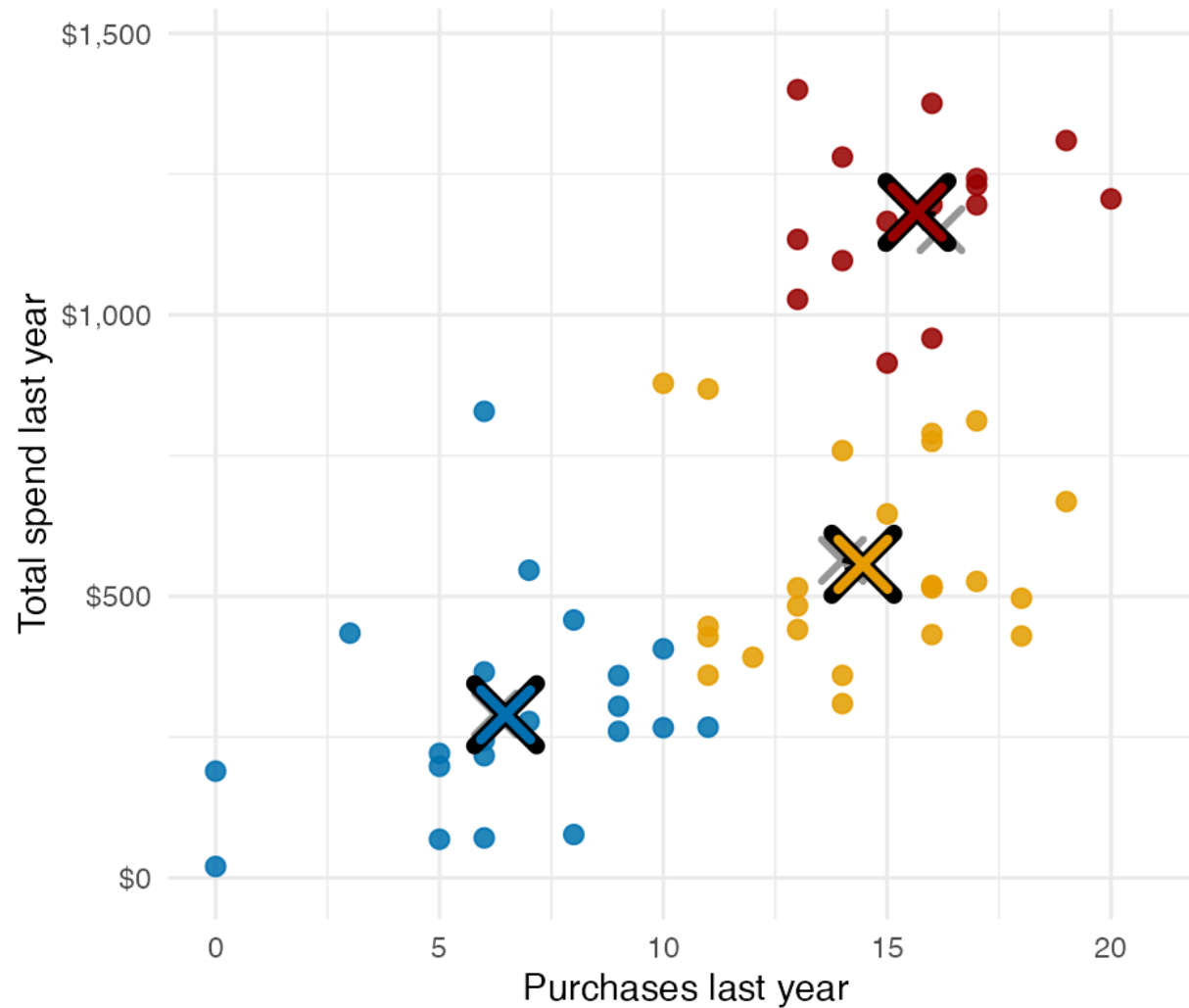
1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

K-means, step by step

Step 3 again: move

each X moves to the middle of its color



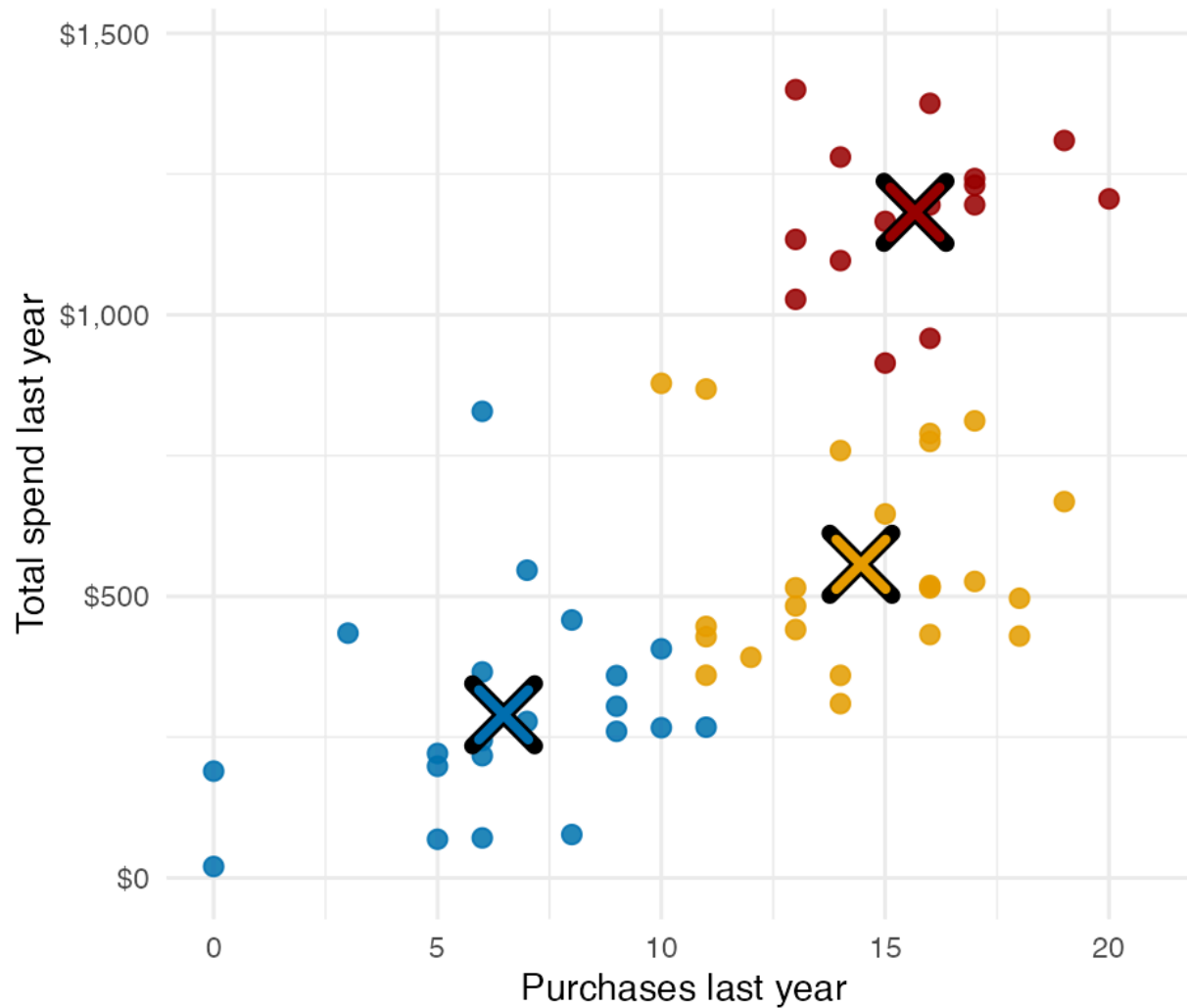
1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

K-means, step by step

Done: nobody switches

k-means stops; same clusters as two slides ago



1. **Drop** 3 centers at random
2. **Assign**: each customer takes the color of its nearest X
3. **Move** each X to the middle of its color
4. **Repeat** 2 and 3 until nobody switches

Each click runs one step. Before you click, guess: which customers will switch color?

Segmenting Office-Store Shoppers

A segmentation, step by step

Where to start, what to do, and why:

Step	What to do	Why
1. Choose the columns	the six importance ratings	they say what customers want , which the store can act on
2. Check the units	standardize if the columns use different scales	k-means measures distance, so big numbers would dominate
3. Choose k	run k-means for k = 2 to 10, look for the elbow	more clusters always fit better; the elbow shows when extra clusters stop paying off
4. Run k-means	k = 3, from many random starts	puts shoppers with similar ratings together
5. Profile	the average rating of each cluster	k-means only says 1, 2, 3; the averages say what each group wants
6. Name	a short name for each cluster	managers act on “bargain hunters”, not on “cluster 1”
7. Describe	age, income, and job of each cluster	who they are, and how to reach them

Step 1 of 7: the data, and which columns

40 respondents rated how important six store attributes are to them when buying office equipment, each on a **1–10** scale:

variety of choice, electronics, furniture, quality of service, low prices, and return policy

- We also know three things about each respondent: **professional** (0/1), **income** (in \$1,000s), and **age**
- We cluster on the **six ratings only**: they say what customers **want**, and that is what the store can change (assortment, prices, service)
- The descriptors say **who** customers are. We keep them out so that income or age cannot decide the groups: we want “price vs. service”, not “young vs. old”. They come back afterwards, to describe each group

Data from R for Marketing Students.

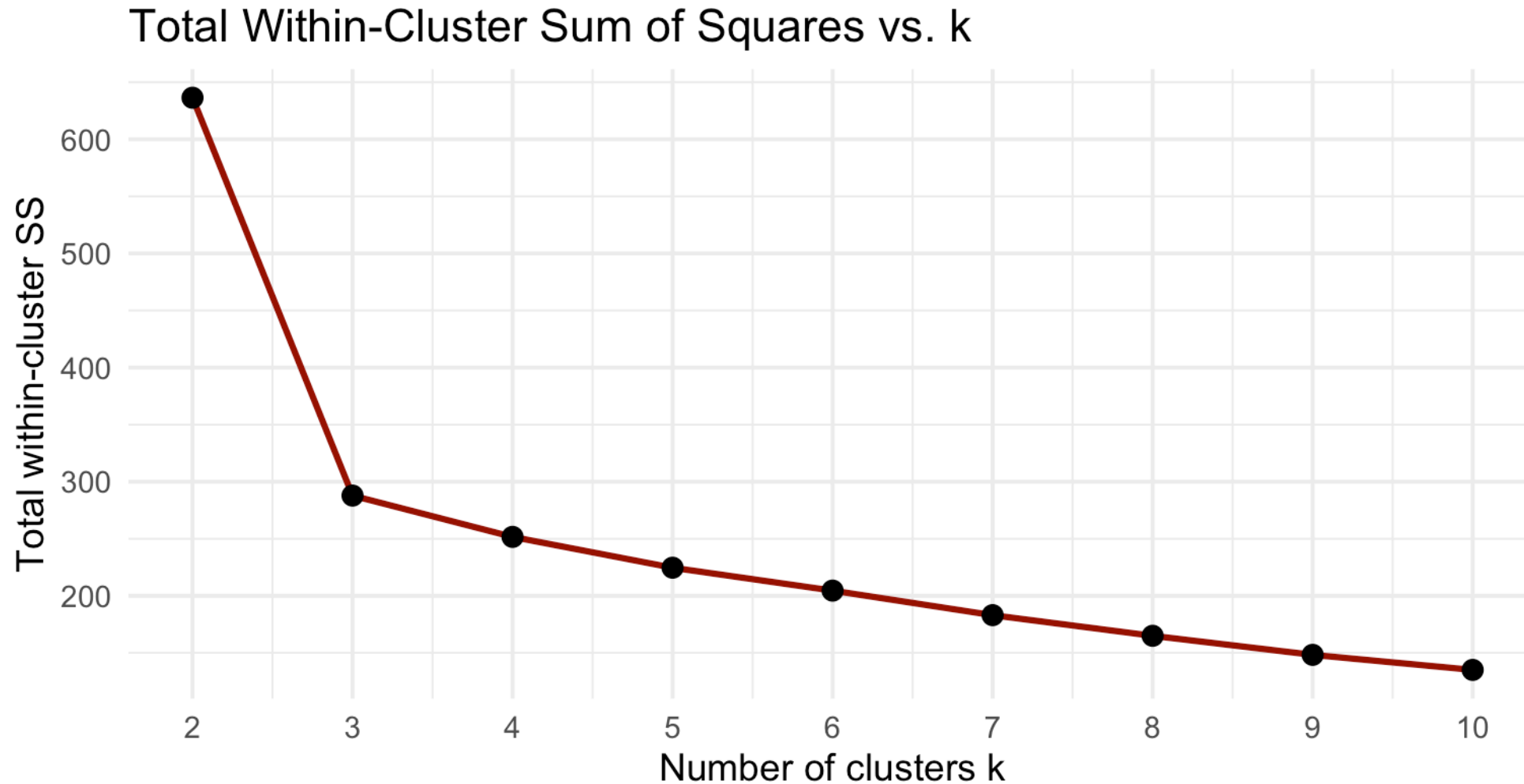
Step 2 of 7: check the units

Suppose you cluster on **income** (\$15,000 to \$70,000) and **age** (20 to 60):

- Two customers \$1,000 apart in income are 1,000 “units” apart; 25 years apart in age, only 25. The distance, and so the clusters, would be **all income and no age**
- The fix: **standardize** first, replacing each column with z-scores (subtract the mean, divide by the standard deviation). Then every column counts equally

Standardize whenever the columns have different units. Today’s survey ratings all share the same 1–10 scale, so we can skip it, and keep the cluster averages in interpretable 1–10 units.

Step 3 of 7: choose k, the elbow curve



For each k, run k-means and add up the squared distances from every point to its cluster center: the “spread left inside the clusters”. Where does the curve bend?

Step 3 of 7: reading the elbow



- Going from 2 to 3 clusters cuts the within-cluster spread from **636 to 288**. From 3 to 4 saves only 36 more, and the curve flattens: the **elbow is at k = 3**
- More clusters always fit better, so “best fit” cannot pick k: with one cluster per customer the fit is perfect and you have learned nothing
- The elbow is a guide, not a law: k must also make **business sense** (three campaigns, not nine)

Steps 4 and 5 of 7: run k-means, profile the clusters

We ran k-means on the six ratings with **k = 3**, the elbow's choice. It split the 40 respondents into clusters of **14**, **18**, and **8**. The labels are just 1, 2, 3, in no particular order. The table shows each cluster's **average rating** of each attribute (1 to 10):

Cluster	Respondents	Variety	Electronics	Furniture	Service	Low prices	Return policy
1	14	6.9	2.8	1.4	3.5	8.3	6.3
2	18	9.1	6.1	5.8	2.4	3.7	3.2
3	8	5.0	4.4	1.8	8.5	2.5	4.4

Read it row by row: what does each cluster rate **high**, and what does it rate **low**? Try to give each row a name before the next slide.

Step 6 of 7: name the segments

We name each cluster by reading **its row in the profile table** (last slide): which attributes it rates **high**, and which it rates **low**.

Cluster (n)	Rates high	Rates low	Our name, and why
1 (n = 14)	low prices 8.3 , return policy 6.3	furniture 1.4, electronics 2.8	bargain hunters : the lowest price, and an easy return if it goes wrong
2 (n = 18)	variety 9.1 , electronics 6.1 , furniture 5.8	service 2.4, low prices 3.7	one-stop shoppers : a huge range, so they can buy everything (electronics and furniture too) in one store
3 (n = 8)	quality of service 8.5	low prices 2.5, furniture 1.8	service seekers : good service matters most, price least

- K-means only says 1, 2, 3. **We** choose the names, by reading the averages
- Check the low ratings too: one-stop shoppers care little about price (3.7), and service seekers care about it least of all (2.5)

Every name must be backed by the averages. If a colleague cannot find your name in the profile table, rename it.

Step 7 of 7: who is in each segment?

The three columns we kept **out** of the clustering describe who landed in each segment:

Segment	Professional	Income	Age
Bargain hunters (14)	50%	\$32,000	31
One-stop shoppers (18)	33%	\$48,000	44
Service seekers (8)	75%	\$48,000	49

- Bargain hunters are the **young, lower-income** group; service seekers are **older professionals**; one-stop shoppers sit in between with the largest headcount
- This is the actionable half: ratings say **what to offer** each segment, descriptors say **who and where they are**

Because these columns played no part in forming the clusters, the differences across segments are a finding, not something we built in.

Six Dimensions on One Screen: PCA

Why PCA?

We already have the clusters and their averages. Why do anything else?

- **We cannot see six ratings.** With two columns (the 60 customers) our eyes could check what k-means did. With six ratings we cannot draw the shoppers, so we cannot check whether the clusters really stand apart
- **Many ratings say the same thing.** Shoppers who care about variety also care about electronics and furniture (furniture and electronics move together: correlation 0.69). Three columns, one idea: **assortment**. PCA finds the few big ideas behind many ratings
- **Managers think in maps.** A two-axis map shows who is close to whom, customers or brands, and where the **gaps** are

PCA turns six ratings into a two-axis **map** that keeps most of the information. We use it to **see** the segments, and to find the few dimensions customers really differ on.

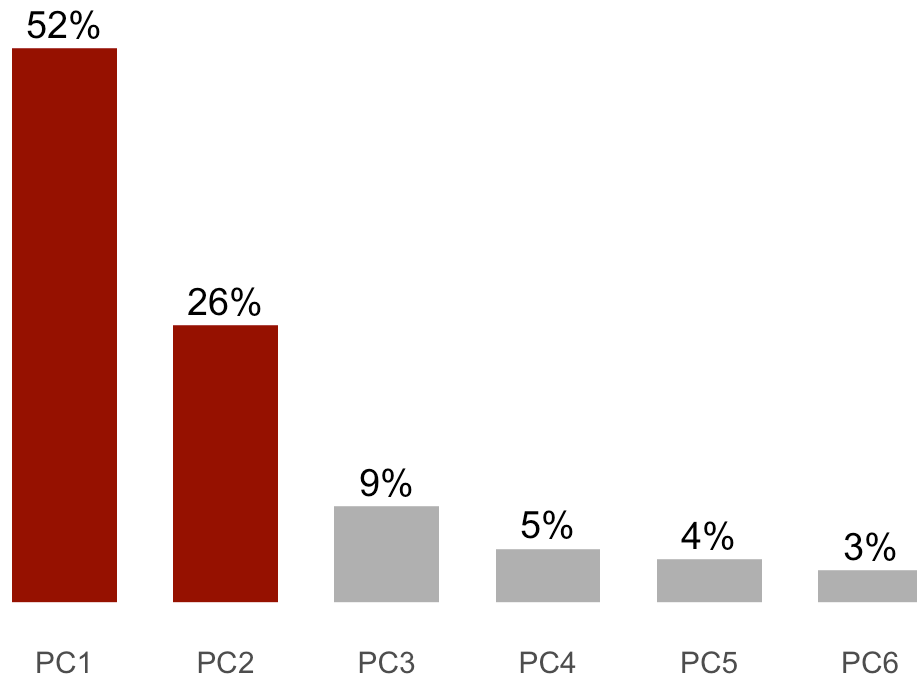
PCA: six ratings, one map

- PCA replaces the six ratings with **new axes**, called **PC1, PC2**, and so on. Each new axis is a **mix** of the six ratings, the way a GPA is a mix of grades
- The axes come **in order**: PC1 captures the biggest differences between shoppers, PC2 the next biggest, and so on
- Keep the first two, and every shopper gets a spot on a **map**

You do not need to know how the computer finds the axes. You need to answer two questions: **how many axes to keep**, and **what each axis means**.

How many axes to keep?

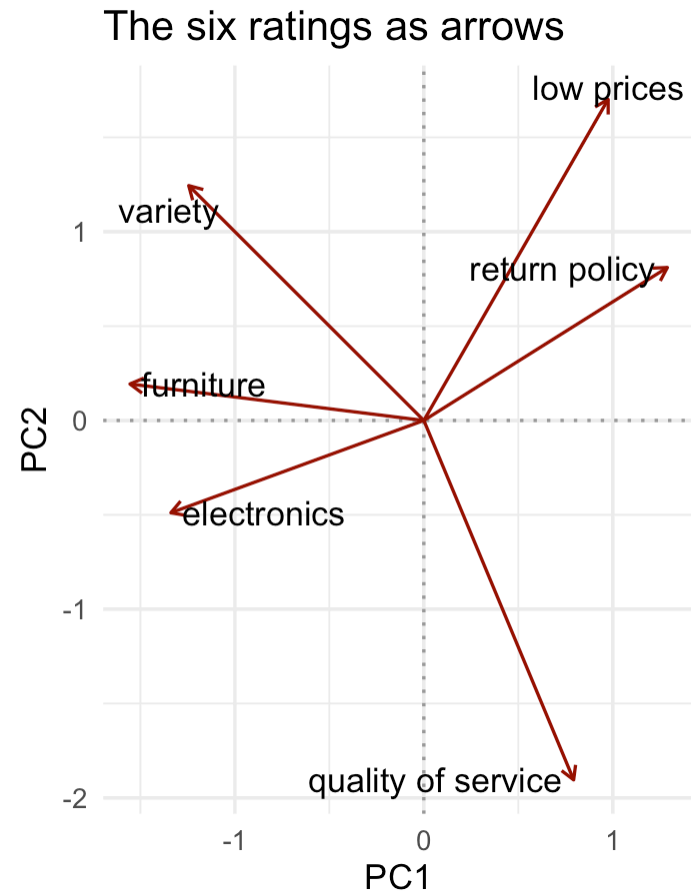
Share of the differences each axis keeps



- Each bar says how much of the differences between shoppers one axis keeps. The six bars add up to 100% (give or take rounding)
- **PC1** keeps **52%** and **PC2 26%**: together **78%**. After PC2 the bars drop to 9% and below
- The rule: keep axes until the bars get **small** (an elbow, like the one in segmentation step 3), and check that the ones you keep add up to **most** of the total

Two axes keep 78% here, so the map tells most of the story. If they kept, say, 40%, the map would hide more than it shows.

What do the two axes mean?



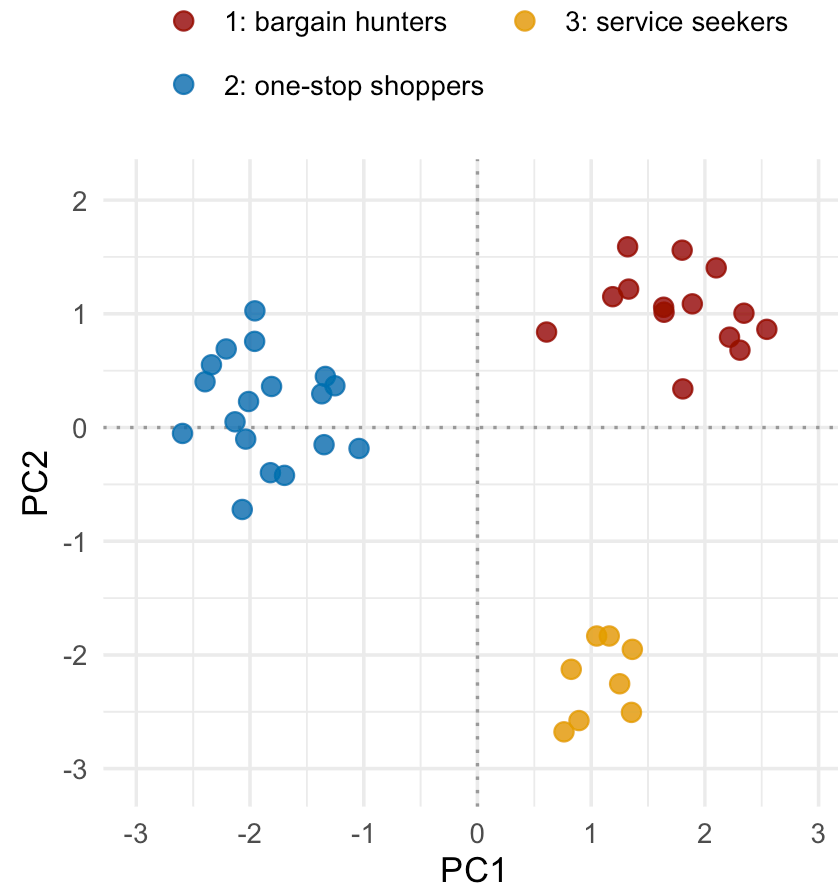
Each arrow is one rating. It points to where shoppers who rate it **high** end up on the map:

- **PC1** (left vs. right): **three** arrows point left: variety, electronics, and furniture. Name the axis by what the three share: left means “wants a **big assortment**”. “Furniture” alone would leave out the other two
- **PC2** (up vs. down): **low prices** reaches highest, **quality of service** reaches lowest. Up means “cares about price, not service”; down, the reverse

PCA could just as well have flipped left and right, or up and down. What matters is which arrows point the same way, and which point opposite ways.

The map: 40 shoppers, colored by cluster

Shoppers on the PCA Map



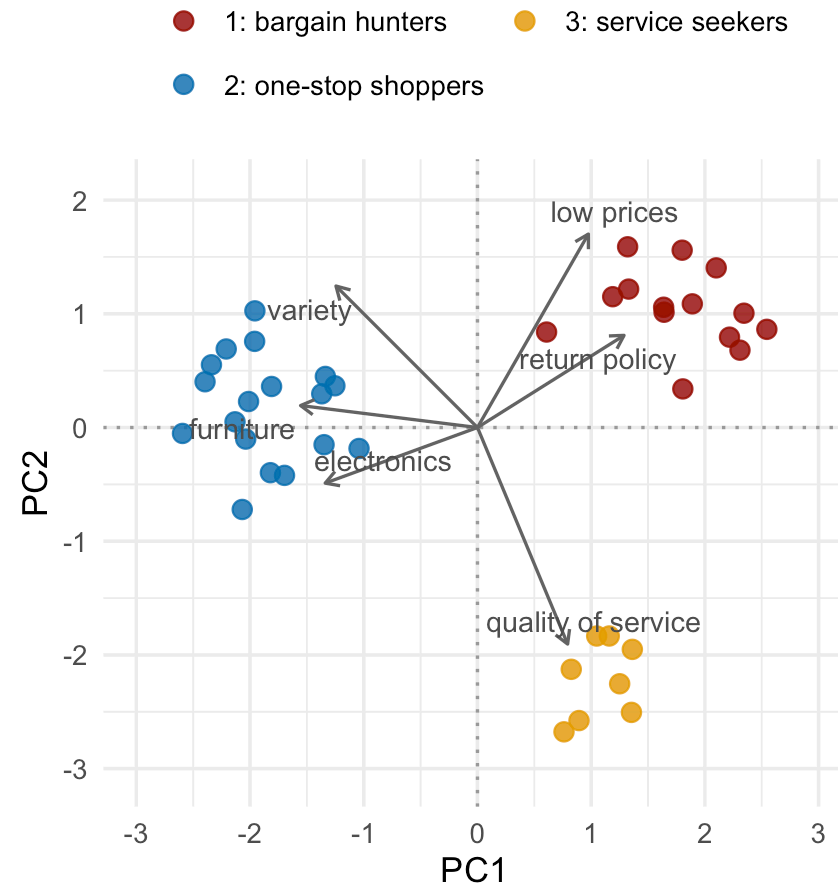
Each dot is a shopper, placed by PCA. The colors are the k-means clusters.

PCA drew the map; k-means picked the colors. The two never talked to each other.

Do the colors form regions?

The map: reading it

Shoppers on the PCA Map



- The clusters sit in **separate regions**, each at the tip of the arrows its profile predicted: bargain hunters by *low prices*, one-stop shoppers by *variety and furniture*, service seekers by *quality of service*
- Why this matters: k-means **always** returns groups (ask for 3, you get 3), even when there are none. PCA never saw the clusters. If the colors still form separate clouds on its map, the groups are **real**. If they were mixed together, k-means would just have cut one blob into pieces

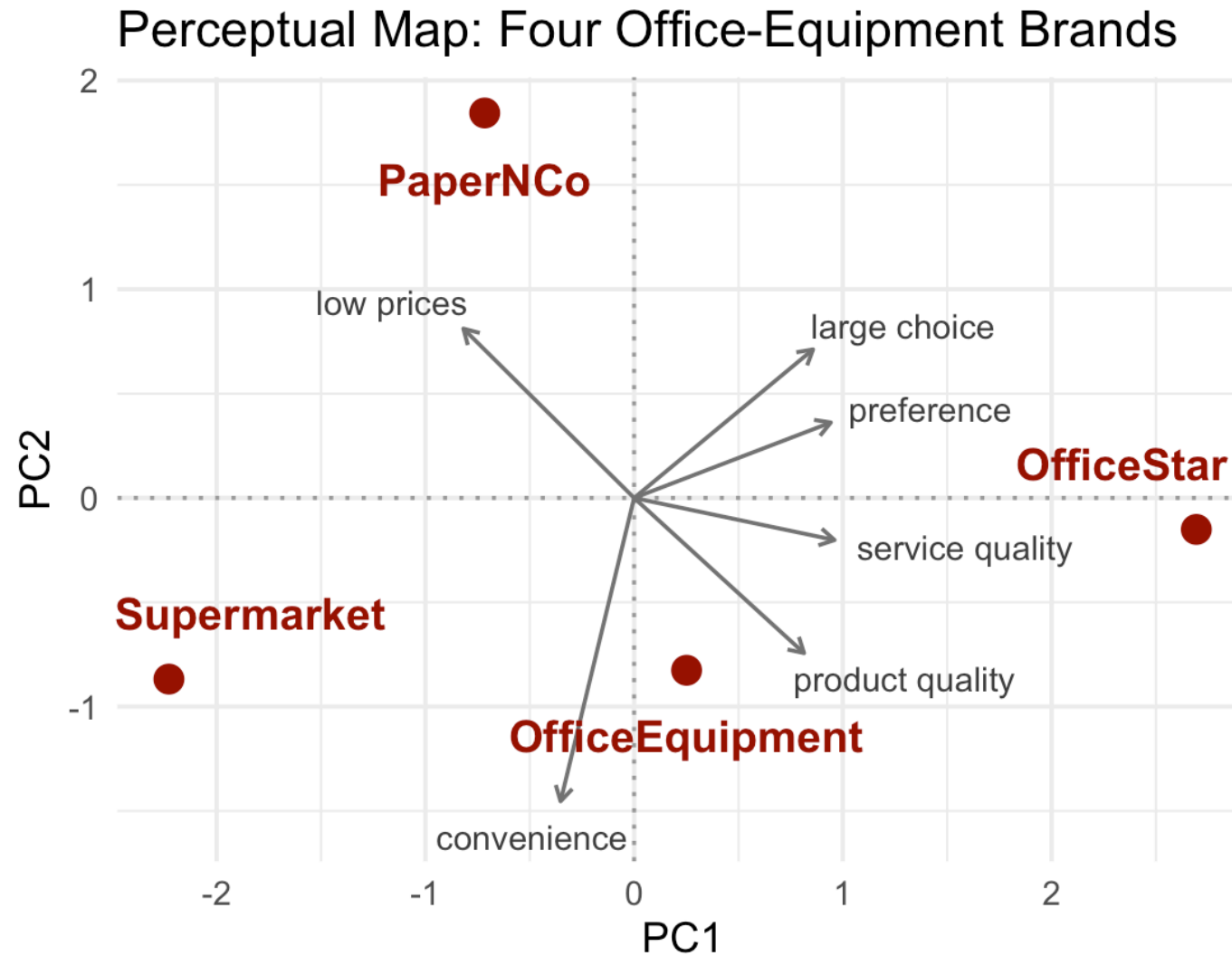
A second dataset: four brands

A **separate survey**. Respondents rated four office-equipment **brands** on six things. Each number is the **average** rating:

Brand	Large choice	Low prices	Service quality	Product quality	Convenience	Preference
OfficeStar	5.2	2.1	4.2	3.7	2.7	5
PaperNCo	4.4	4.5	2.3	2.6	1.4	3
OfficeEquipment	3.9	2.6	3.1	3.1	4.7	3
Supermarket	2.3	4.1	1.8	2.9	5.1	1

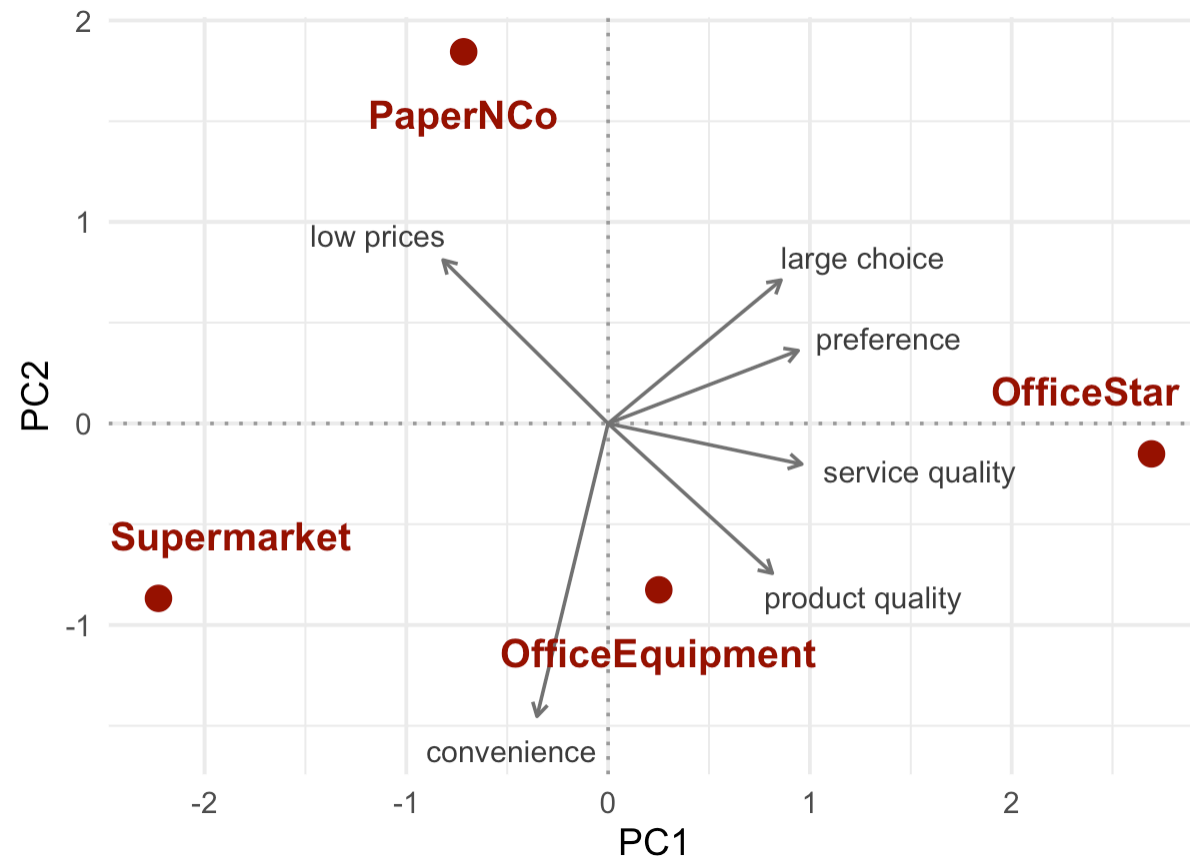
- **Rows are brands** now, not shoppers. Same problem as before: six columns, and a screen has two axes
- **Preference** is how much respondents like the brand overall
- Same PCA, read the same way: which brands sit close together, and which arrows point at each brand?

Four brands on one map: a perceptual map



Each red dot is a **brand**; each arrow is one of the six ratings. Which brands compete head-to-head? Who owns which attribute?

The perceptual map: reading it



- **OfficeStar** sits alone on the right, next to *service quality*, *product quality*, *large choice*, and **preference**: the strong brand people like best
- **Supermarket** is its mirror image, on the left by *low prices* and *convenience*. **PaperNCo** owns the cheap corner; **OfficeEquipment** leans on convenience, with no clear strength
- Two axes keep **98%** of the differences between the brands: the map is almost the whole story

Perceptual maps answer positioning questions: who are my real competitors (neighbors on the map), what do I own (my nearest arrows), and where is the **gap** nobody occupies.

Clustering and PCA, side by side

	K-means	PCA
Question	which customers belong together?	how do I show six ratings on one map?
Output	a group label per customer (1, 2, 3)	two scores (PC1, PC2) per customer
You choose	the columns, and k	the columns, and how many axes to keep
You read it with	the averages (profile table)	the arrows
Marketing use	segmentation	perceptual maps; seeing the segments

Three cautions for both methods:

- **Standardize** when the columns use different units
- Clusters and axes mean **nothing** until you read them: the averages for clusters, the arrows for axes
- Clusters **describe** customers; they do not explain them. Service seekers are older on average, but that does not mean age makes people want service. It only tells us who tends to be in the group

From clustering to PCA: the steps, and why

Step	What to do	Why
1. Pick the columns	what customers want (the ratings); standardize if the scales differ	groups built on needs, with fair distances
2. Find the groups	choose k with the elbow, then run k-means	shoppers with similar ratings end up together
3. Read the means	the average rating of each cluster, then name each one	what each group wants, in exact numbers
4. Describe	age, income, and job of each cluster	who they are, and how to reach them
5. Check the axes	run PCA; keep two axes if they hold most of the differences	a map is only useful if it tells most of the story
6. Map and compare	plot the shoppers on PC1 and PC2, color them by cluster, add the arrows	see whether the groups really separate, and what separates them

Steps 3 and 6 **check each other**. The means give exact numbers, one attribute at a time; the map shows all the shoppers at once. The check: each cluster should sit next to the arrows of what it rates **highest**. Here it does (bargain hunters, who rate low prices 8.3, sit by the low-prices arrow), so the segments are real, not something k-means made up.

Wrap-up

The R code and data behind every chart and table:

- [w6-code.zip](#): both scripts and both datasets
- [w6-1-clustering-class.R](#): elbow, k-means, profiles, the PCA map
- [w6-1-pca-class.R](#): the brand perceptual map

Required reading: chapters [5](#), [6](#), and [7](#) of *R for Marketing Students*. Optional: chapters 11.1–11.3 of *R for Marketing Research and Analytics*. Thursday: recommendation systems.

Questions?